

Week 8&9

SCENARIO

Imagine you are building a simple desktop application for yourself.

Instead of writing daily tasks on paper, you decide to build a **Java application** that allows you to:

- Add new tasks
- View all tasks
- Save tasks into a file
- Load previous tasks automatically

Every time the application starts, it should remember the tasks saved previously.

INSTRUCTION

1. Login to your GitHub & select your OOP repository.
2. Create a new directory called "week_8-9".
3. Move into the week_8-9 directory & create the following java files:
3.1. Main.java

ACTIVITY

1. Create an *ArrayList* & allow the users to enter THREE (3) to-do list & display them in the terminal
2. Enhance the program by:
 - 2.1. Saving all tasks into task.txt
 - 2.2. Reading the contents of task.txt
 - 2.3. Displaying the tasks that were loaded from the file.

```
import java.util.ArrayList;
import java.io.*;

public class Main {

    public static void main(String[] args) {

        // code missing;

        // code missing;

        System.out.println("===== ADD TASKS =====");

        // Allow user to enter 3 tasks
        for (int i = 0; i < 3; i++) {

            System.out.print("Enter Task " + i + ": ");

            String task = "";

            while (task.isEmpty()) {

            }

            // Display tasks
            System.out.println("\n===== TASK LIST =====");

            for (int i = 0; i < tasks.size(); i++) {

                System.out.println((i + 1) + ". " + tasks.get(i));

            }

        }

    }

    // Saving the task to a text file
    try (BufferedWriter writer =
        new BufferedWriter(new FileWriter("task.txt"))) {

        for (String task : tasks) {

            writer.write(task);
            writer.newLine();

        }

        System.out.println("\nTasks saved successfully.");

    } catch (IOException e) {

        System.out.println(e.getMessage());

    }

    // Read from file
    System.out.println("\n===== TASKS LOADED FROM FILE =====");

    try (BufferedReader reader =
        new BufferedReader(new FileReader("task.txt"))) {

        String line;

        while ((line = reader.readLine()) != null) {

            System.out.println(line);

        }

    } catch (IOException e) {

        System.out.println(e.getMessage());

    }

}
```

Activity (1) - EXPECTED OUTPUT

```
===== ADD TASKS =====
Enter Task 1: drink water
Enter Task 2: completing AI course
Enter Task 3: drink water again

===== TASK LIST =====
1. drink water
2. completing AI course
3. drink water again
```

Activity (2) - EXPECTED OUTPUT

```
===== ADD TASKS =====
Enter Task 1: drink some more water
Enter Task 2: walking at the park
Enter Task 3: eat healthy snack

Tasks saved successfully.

===== TASKS LOADED FROM FILE =====
drink some more water
walking at the park
eat healthy snack
```